

Article

# Exposure to Online Alcohol Marketing and Adolescents' Drinking: A Cross-sectional Study in Four European Countries

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## Abstract

**Aims:** The Internet is the leading medium among European adolescents in contemporary times; even more time is spent on the Internet than watching television. This study investigates associations between online alcohol marketing exposure and onset of drinking and binge drinking among adolescents in four European countries.

**Method:** A total of 9038 students with a mean age of 14.05 (SD 0.82) participated in a school-based survey in Germany, Italy, the Netherlands and Poland. Logistic regression analyses of cross-sectional cross-country survey data were undertaken. Exposure to online alcohol marketing, televised alcohol advertising and ownership of alcohol-branded items was estimated to be controlled for relevant confounders. Onset of drinking and binge drinking in the past 30 days were included in the study as outcome variables.

**Results:** Adjusted for relevant confounders, higher exposure to (online) alcohol marketing exposure was found to be related to the odds of starting to drink ( $p < 0.001$ ) and the odds of binge drinking in the past 30 days ( $p < 0.001$ ). This effect was found to be consistent in all four countries. Active engagement with online alcohol marketing was found to interact more strongly with drinking outcomes than passive exposure to online alcohol marketing.

**Conclusions:** Youngsters in the four European countries report frequent exposure to online alcohol marketing. The association between this exposure and adolescents' drinking was robust and seems consistent across national contexts.

## INTRODUCTION

### Underage alcohol consumption in the European union

In Europe, alcohol is the third leading risk factor for disease and mortality after tobacco and high blood pressure (World Health Organization, 2009). The adolescent brain is particularly susceptible to alcohol as,

besides direct consequences, alcohol consumption during adolescence is associated with alcohol problems later in life (McCambridge *et al.*, 2011). Furthermore, alcohol use among European youth is highly prevalent: 87% of 15 and 16 year olds have drunk alcohol at least once in their lifetime, and 57% in the past 30 days (Hibell *et al.*, 2012).

## The role of alcohol advertising

The role of alcohol advertising in influencing adolescents to begin drinking has gained growing attention. In general, longitudinal studies show a moderate but significant association between alcohol advertising exposure and adolescents' drinking (Anderson *et al.*, 2009; Smith and Foxcroft, 2009). Studies, in which online alcohol marketing is part of a cumulative exposure measure, suggest a positive association between exposure to alcohol marketing and young people's drinking behavior (Gordon *et al.*, 2011a; Pinsky *et al.*, 2010; Tucker *et al.*, 2013). A study by Lin *et al.* (2011) underlines the importance of examining the impact of web-based marketing in addition to traditional marketing, which has found to be a significant predictor of onset of drinking and amount of drinking among teenagers from New Zealand. Additionally, results of a cross-sectional study among Australian adolescents are consistent with studies from other countries and suggest that exposure to online alcohol advertisements is associated with adolescent drinking patterns (Jones and Magee, 2011). The only study conducted in Europe, which included exposure to nontraditional media, suggests a cumulative effect of alcohol marketing channels on drinking by Scottish youth (Gordon *et al.*, 2011a, 2011b). Recently, researchers have addressed the need for more survey data on the behavioral effects of youth exposure to and participation in online alcohol marketing (Jernigan and Rushman, 2014).

## Alcohol advertising in digital media in Europa

This study looks at the association between alcohol marketing exposure and initiation of alcohol use, recent binge drinking and volume of alcohol consumption among European youth, with a special focus on the influence of alcohol marketing in digital media. This is particularly relevant due to the large amounts of time European youngsters spend using digital media. The Internet is the leading medium in contemporary times; even more time is spent on the Internet than watching television (European Interactive Advertising Association, 2007). The alcohol industry utilizes the Internet as a marketing channel, most notably via producers' websites, by banners on other websites and on social networking sites (Nicholls, 2012).

As early as 2004, the Center on Alcohol Marketing and Youth (CAMY, 2004) raised concern about the fact that nearly 700,000 in-depth visits of underage persons to 55 alcohol websites were recorded during the last 6 months of 2003. Nowadays, the alcohol industry has increased its attention to social media channels. For example, as of November 2011, 10 alcohol brands had uploaded 35,725 photos on Facebook (CAMY, 2012).

A study on alcohol advertisements of five UK alcohol brands indicated that all five of the brands maintained a Facebook page, Twitter page and YouTube channel, with varying levels of user engagement (Winpenny *et al.*, 2014). Facebook pages could not be accessed by an under aged user, but in most cases, YouTube content and Twitter content could be accessed by all ages. Meanwhile, a survey from 2011 indicated that half of the Irish 9 to 12 year olds who are active on Facebook report lying about their age to get access to pages on Facebook (O'Neill, 2011).

Barry *et al.* (2015) used fictitious YouTube user profiles of a 14, 17 and 19 year olds to view brewer-sponsored YouTube channels of 16 popular beer brands. They discovered that every underage profile, regardless of age, was able to successfully subscribe to each of the official YouTube channels. On average, two-thirds of the brands' channels were successfully viewed.

There is little data available on alcohol marketing in the four European countries studied. The countries differ with respect to alcohol advertising expenditure per capita. The alcohol advertising expenditures per capita of measured media (television, radio, magazines/newspapers and billboards) in 2007 were respectively: €6.70 in Germany; €5.15 in the Netherlands and €2.86 in Italy. There is a limited number of publicly available figures for Poland. In Poland, televised alcohol advertising expenditures have almost doubled in the five years since the television ban on beer advertising was lifted, growing from €46 million in 2002 (Carat, 2005) to €96 million in 2007 (TNS OBOP, 2009). With alcohol advertising expenditures on television at €3.6 per capita, expenditures in Poland are still relatively low compared to those of other large European countries (EACA, 2009). Regulations on alcohol marketing may explain why. In Poland, there is a ban on advertising of spirits and wine, while in Germany and in the Netherlands, alcohol marketing is mostly self-regulated. In Italy, there are several legal restrictions in place, but in practice, the self-regulation codes that regulate the content of alcohol advertising are the only restrictions that are (limitedly) enforced (Beccaria, 2001; de Bruijn *et al.*, 2012).

## Objectives of the study

A study in Germany, Italy, the Netherlands and Poland reports the findings of the first cross-national survey that examines the association of alcohol marketing exposure and adolescents' drinking habits. The article builds on preliminary findings from the AMPHORA project (de Bruijn, 2012) and focuses on the role alcohol marketing in digital media plays on influencing adolescents to begin drinking. Countries selected are geographically distributed over Europe and differ with respect to drinking cultures and alcohol marketing regulations (de Bruijn, 2011; Hibell *et al.*, 2012).

## METHOD

### Survey design

A draft of the survey was pretested with a sample of 100 students in each country. In each country, data were collected through self-administered online questionnaires which were anonymous. Students who volunteered to participate gave active consent. In Germany, active parental consent was required, and in all other countries, passive parental consent was used. Ethical approval for the study was gained by the European Commission and the Ethical Board of the Radboud University (number ECG 24092009).

### Participants

In each country, a stratified sample of schools in an urban and rural area was taken, giving a total of 339 schools, of which 163 were eligible for participation. Schools aimed at students with special needs were excluded from the sample. Eligible schools had sufficient computers available to fill out the computerized questionnaire and had a school director who agreed to participate in the study. Of the eligible 10,810 students who were invited, 9709 participated (89.8% response rate). Two-thirds of the non-response of individuals was due to lack of active parental consent ( $N = 732$ ). Technical difficulties concerning the Internet connection at the times of the survey, or other technical failures, reduced the sample to 9032 students. Of those who ultimately participated, responses were recorded using the online questionnaire. The mean age of the sample was 14.05 ( $SD = 0.82$ ), and the sample was divided by sex (50% were male).

## Measures

### Alcohol use

Nondrinkers were identified by asking respondents, 'When was the last day you drank alcohol?'. Those who answered 'I never drink alcohol' were categorized as nondrinkers ( $n = 4553$ ), all other were categorized as drinkers ( $n = 4485$ ). Binge drinking was established by asking respondents, 'During the last 30 days, how many times did you have five or more drinks on the same occasion?'. Students who reported 'never' were classified as 'not a binge drinker', all others were classified as 'a binge drinker'. Drinking questions mirrored questions used in the ESPAD survey (Hibell *et al.*, 2012).

### Alcohol marketing exposure

Dichotomous questions used by Gordon *et al.* (2011a) were adapted to measure the frequency of exposure to alcohol marketing in online media with a 5-point Likert scale (1, never; 2, rarely; 3, sometimes; 4, often and 5, very often). Respondents were asked whether they had ever received 'promotional mail, emails or joke, chain or wind up emails mentioning alcohol brands', 'looked at a website for alcohol brands or about drinking (does not include health-related sites)', 'downloaded a mobile phone or computer screensaver containing an alcohol brand name or logo', 'used a profile page on sites such as Hyves, Facebook, MSN or Myspace containing an alcohol brand or logo' and 'noticed an Internet page that contained an alcohol advertisement'. All items were combined into one factor (Eigen value = 2.544 with 50.88% variance explained and Cronbach's alpha = 0.742).

Ownership of an alcohol-branded promotional item was determined by asking respondents 'Do you own an item—like a t-shirt, lighter, matches, hat, or sunglasses—with an alcohol brand name on it?' (Henriksen *et al.*, 2008). Those who answered 'yes' to this question were coded as being an owner of an alcohol-branded promotional item. Respondents answering 'no' or 'I don't know' were coded as not being an owner of an alcohol-branded promotional item.

Exposure to televised alcohol advertising was measured by asking respondents about the frequency with which they had seen a selection of eight television programs in February 2010 (~1–2 months before the survey). Respondents indicated their frequency of watching with a 5-point Likert scale (1, never; 2, rarely; 3, sometimes; 4, often and 5 very often), which was recoded into a score from 0 to 4. The list of television programs in each country questionnaire was drawn from a list of most popular television programs among 13 to 17 year olds in each country in which (or right before or after) an alcohol commercial was aired. A total score of televised alcohol advertising exposure was calculated by dividing the number of advertisements broadcasted in each program by the total number of alcohol advertisements broadcasted in all eight television programs in the country. This score was multiplied by the frequency of watching the program divided by four. These scores were summed for each respondent to obtain a score between zero and one for respondents in all four countries.

### Confounders

Demographic data were recorded for age, gender, education and smoking behavior (yes/no). Social influences were measured by perceived constraints toward alcohol use in religion (yes/no), number of four closest friends drinking and their perceived approval to drink, alcohol use of mother (Kerr *et al.*, 2012) and her perceived approval to drink. Additionally, nonalcohol-branded media exposure measures were included. Internet use was measured by asking

respondents 'On a usual school day (Monday to Friday), how many hours do you spend on using the Internet?' indicating (1) none; (2) <1 hour; (3) 1–2 hours; (4) 3–4 hours and (5) 5 hours or more. Additionally, exposure to nonalcohol-branded television programs was measured by asking respondents the frequency of watching four television programs in which (or right before or after) no alcohol advertisements were aired. Selection of television programs in each country was based on a list of most popular television programs among 13 to 17 year olds in each country in September 2010, obtained from Nielsen Media.

## Statistical analysis

Analyses were conducted with M-Plus version 6.1 (Muthén and Muthén, 2010). Data are nested due to the school-based sample design and cross-country setting. We were not able to estimate a three-level multilevel model due to the limited number of countries (Maas and Hox, 2005). Alternatively, in all models, class was identified as a cluster variable that resulted in a sandwich adjusted variance (Hanewinkel *et al.*, 2012) and country differences were taken into consideration.

Descriptive analysis was carried out by observing the prevalence of drinking and other measures in each country compared to the total sample. Univariate and multivariate logistic regression models were carried out to examine the interaction of online alcohol marketing exposure items and drinking outcomes, taking into account relevant confounders.

Binary logistic regression was carried out to observe whether alcohol marketing exposure and more specifically online alcohol marketing exposure was associated with being a nondrinker versus a drinker independent of potential confounders. In a second step, possible differences in effect sizes of the impact of online alcohol marketing were examined by unconstraining the effect sizes to be equal between countries with all other parameters equal. In a third step, models are run for each country separately. The same model of the total sample was then run replacing onset of drinking with binge drinking.

## RESULTS

The table in S1 presents the distributions of the study variables among respondents of the different countries and the total sample (see also S2 for more details). Nondrinkers accounted for approximately half of the respondents, and were lowest in Italy with approximately one-third of the sample having never used alcohol. Amounts of alcohol consumed and type of product consumed differed between countries. Approximately three-fourth of the respondents were not classified as binge drinkers (in the last 30 days) (65% in Italy, 70% in Germany and 82% in the Netherlands and Poland).

Over one-fourth of the respondents received promotional emails containing alcohol advertisements, and one in five looked at websites for alcohol brands. One-third of the respondents used a profile website on social media with an alcohol advertisement, ranging from one-fifth in Germany to half of the respondents in Poland. Approximately two-thirds of the respondents noticed an alcohol advertisement on an Internet page ranging from half of the respondents in Germany to more than three-fourths of the respondents in Poland. A smaller proportion of respondents downloaded a screensaver containing an alcohol advertisement on their computer or mobile phone ranging from 29% in Italy, to 13% in Poland.

Zero-order correlations showed significant associations between alcohol drinking measures and all survey variables. All alcohol advertising items were found to be correlated to each other and to media use measures. See S3 for details.

Table 1 presents partial associations between type of online alcohol marketing exposure and drinking outcomes. When tested univariately, exposure to each type of online alcohol marketing was associated with an increase in the likelihood of beginning to drink (onset of drinking) and past 30 days binge drinking. This association was strongest for measures which involved an active role of the adolescent (looking at a website for alcohol brands and downloading an alcohol-branded screensaver).

With the sizeable correlations between the predictors, the unique variance explained by each of the variables was relatively low. When taking into account the impact of other online alcohol marketing exposure measures, receiving alcohol-branded promotional emails explains no unique variance of onset of drinking. Noticing online alcohol advertisements increased the risk of binge drinking in the past month. However, when taking into account the impact of

exposure to other types of online alcohol marketing, noticing online alcohol advertisements no longer increased the risk of binge drinking in the past month. Inspection of the odds ratios suggests that the predictors were indicators of the underlying (latent) variable described by the model, which can be interpreted as online alcohol marketing exposure.

Table 2 presents binary logistic regression odds of onset of drinking (see S4 for more details). Higher exposure to *online alcohol marketing* was found to be associated with increased odds of being a drinker, while adjusting for all the listed confounders. Also, a positive effect was found for *ownership of alcohol-branded items* and higher *exposure to televised alcohol advertising*. Similar results were found for binge drinking.

We found few between-country differences in the strength of the adjusted relationship between online alcohol marketing exposure and onset of drinking. When estimating a model in which only the coefficients of the marketing exposures were estimated freely for each country (and the coefficients of the covariates are constrained to be equal), a Wald test showed that the effect sizes of the impact

**Table 1.** Partial associations between online alcohol marketing exposure measures and onset of drinking/past 30 days binge drinking tested univariate and multivariate

|                                        | Onset of drinking |             |              |             | Past 30 days binge drinking |             |              |             |
|----------------------------------------|-------------------|-------------|--------------|-------------|-----------------------------|-------------|--------------|-------------|
|                                        | Univariate        |             | Multivariate |             | Univariate                  |             | Multivariate |             |
|                                        | OR                | CI          | OR           | CI          | OR                          | CI          | OR           | CI          |
| Received promotional emails            | 1.10***           | (1.07–1.13) | 1.00         | (0.97–1.03) | 1.15***                     | (1.11–1.18) | 1.05*        | (1.02–1.09) |
| Looked at a website for alcohol brands | 1.24***           | (1.20–1.29) | 1.14***      | (1.09–1.19) | 1.23***                     | (1.19–1.27) | 1.11***      | (1.07–1.16) |
| Downloaded alcohol-branded screensaver | 1.23***           | (1.18–1.28) | 1.12***      | (1.08–1.17) | 1.22***                     | (1.18–1.26) | 1.12***      | (1.08–1.16) |
| Used alcohol-branded social media page | 1.13***           | (1.10–1.17) | 1.06*        | (1.03–1.10) | 1.16***                     | (1.13–1.19) | 1.06**       | (1.03–1.10) |
| Noticed online alcohol advertisement   | 1.08***           | (1.08–1.13) | 1.06***      | (1.03–1.09) | 1.08**                      | (1.05–1.10) | 1.00         | (0.98–1.03) |

Notes: Reported ORs are controlled for gender, smoking, age, education level, religious constraints against alcohol, alcohol use peers, alcohol use mother, peer permission to drink and maternal permission to drink. OR, odds ratio; CI, 95% confidence interval.

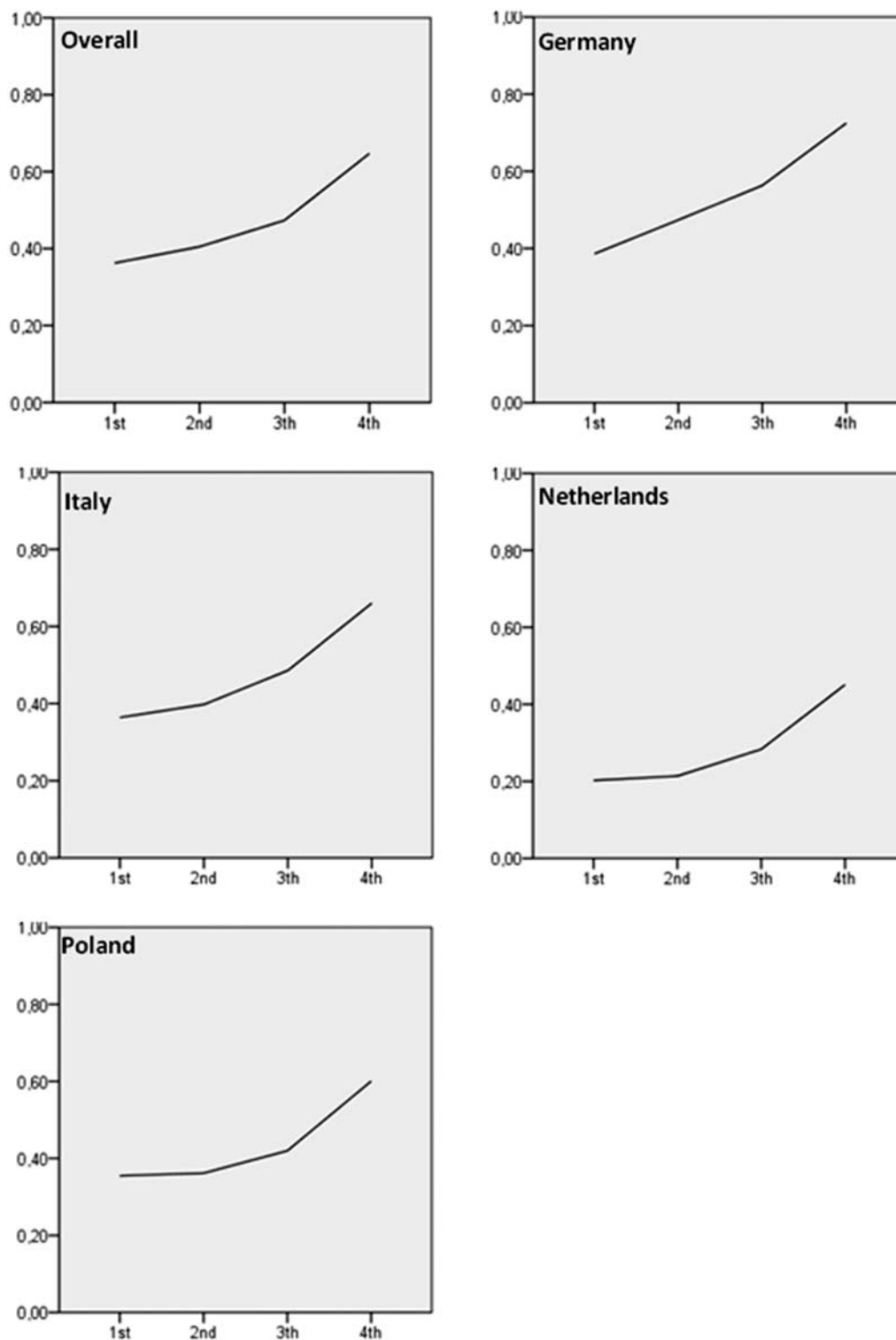
\*\*\* $p < 0.001$ ; \*\* $p < 0.01$ ; \* $p < 0.05$ .

**Table 2.** Binary logistic regression results on the odds of onset of drinking and onset of alcohol use

|                                    | Onset of drinking |             | Binge drinking |             |
|------------------------------------|-------------------|-------------|----------------|-------------|
|                                    | OR                | (95% CI)    | OR             | (95% CI)    |
| Male                               | 1.08*             | (0.99–1.18) | 1.04           | (0.94–1.16) |
| Age                                | 1.13***           | (1.06–1.20) | 1.17***        | (1.10–1.24) |
| Education                          | 1.02              | (0.96–1.08) | 0.97           | (0.91–1.04) |
| Smoking                            | 2.02***           | (1.84–2.21) | 1.74***        | (1.61–1.89) |
| Restrictions religion              | 0.97              | (0.93–1.01) | 1.01           | (0.97–1.06) |
| Alcohol use peers                  | 1.31***           | (1.26–1.35) | 1.31***        | (1.27–1.36) |
| Alcohol use mother                 | 1.17***           | (1.12–1.23) | 1.12***        | (1.07–1.17) |
| Permission peers                   | 1.42***           | (1.35–1.49) | 1.27***        | (1.20–1.34) |
| Permission mother                  | 0.82**            | (0.69–0.98) | 0.87*          | (0.75–1.02) |
| Internet use                       | 1.06***           | (1.03–1.10) | 1.09***        | (1.05–1.14) |
| TV nonalcohol advertising exposure | 0.99              | (0.93–1.05) | 1.03           | (0.97–1.09) |
| TV alcohol advertising exposure    | 1.21*             | (0.97–1.50) | 1.20*          | (0.95–1.52) |
| Ownership alcohol branded item     | 1.29***           | (1.17–1.42) | 1.15***        | (1.05–1.26) |
| Online alcohol ad exposure         | 1.26***           | (1.17–1.36) | 1.24***        | (1.16–1.32) |
| N                                  | 8985              |             | 8982           |             |
| R <sup>2</sup>                     | 0.60              |             | 0.52           |             |
| CFI                                | 1.00              |             | 1.00           |             |
| RMSEA                              | 0.00              |             | 0.00           |             |

Note: OR, odds ratio; CI, 95% confidence interval.

\*\*\* $p < 0.001$ ; \*\* $p < 0.01$ ; \* $p < 0.05$ .



**Fig. 1.** Predicted probability of onset of drinking (Y-axis) by different levels of exposure to online alcohol marketing (X-axis in quartiles), overall and by country. Adjusted regression coefficients (95% confidence interval between brackets) with \* $p < 0.01$ ; \*\* $p < 0.001$ : Four countries:  $\beta$  (adj.) = 0.233(0.159–0.307)\*\* ( $N = 8985$ ); Germany:  $\beta$  (adj.) = 0.307(0.152–0.463)\*\* ( $N = 1857$ ); Italy:  $\beta$  (adj.) = 0.260(0.114–0.407)\*\* ( $N = 2654$ ); The Netherlands:  $\beta$  (adj.) = 0.251(0.082–0.421)\*\* ( $N = 2038$ ); Poland:  $\beta$  (adj.) = 0.193(0.047–0.340)\* ( $N = 2436$ ).

of exposure to online alcohol advertising on onset of drinking do not differ significantly between countries (Wald (df 3) = 2.152, with  $p = 0.541$ ). Also, this model has a poorer fit (CFI = 0.94; RMSEA = 0.35) than the model in which all parameters in all countries were constrained to be equal (CFI = 1.00; RMSEA = 0.00). Figure 1 shows standardized estimates for the relationship between online alcohol marketing exposure (expressed in quartiles) and predicted probability of onset of drinking, adjusted for all confounders, overall and by country. The graphs indicate a dose-response relationship between online alcohol marketing exposure and probability of starting to drink alcohol ( $p < 0.01$  in overall and country models).

## DISCUSSION

To the best of our knowledge, this is the first study that has examined the impact of different levels of frequency of exposure to (online) alcohol marketing practices and the first cross-country study that examined the impact of alcohol advertising on adolescents' drinking behavior. The results show that frequency of exposure to alcohol marketing was found to be associated with drinking behavior, even when controlled for media use like hours spent using the Internet, demographic and social factors (alcohol use and perceived approval of peers and mother). Not surprisingly, active engagement with online alcohol marketing (e.g. by downloading an alcohol-branded screensaver on the computer or mobile phone) had the strongest interaction with drinking outcomes. Not only was the association between online alcohol marketing exposure and drinking outcomes statistically significant, its effect size was also substantial when compared to the impact of other factors. The association of exposure to online alcohol marketing and alcohol use (onset drinking, past 30 days binge drinking and the volume of consumption among drinkers) was found to be consistent in all examined countries. These findings are in line with other studies that studied the impact of exposure to online alcohol marketing on adolescents' drinking (Gordon *et al.*, 2011a, 2011b; Jones and Magee, 2011; Lin *et al.*, 2011). However, it is not just commercial advertising messages that matter: recent studies found a strong association between young people explicitly presenting themselves as drinkers (assuming an 'alcohol identity') on social networking sites and harmful drinking behavior (Moreno *et al.*, 2012; Ridout *et al.*, 2012). This illustrates the problem of the lines being blurred between commercial advertising messages and user-generated content on social media sites (Moreno *et al.*, 2012; Ridout *et al.*, 2012).

The frequency of exposure to online alcohol marketing among respondents indicates the degree in which alcohol producers are able to reach young people at a very vulnerable age. For example, Heineken and Google have started a global partnership which increases the international beer producer's YouTube activity. This deal very likely means that at least 103 million minors around the world are being exposed to the harmful effects of alcohol marketing on a monthly basis (EUCAM, 2011). Like in most countries, the volume of alcohol marketing on the Internet is not regulated by law in the countries examined. These countries rely on self-regulation which seems unable to protect young people against high exposure to alcohol marketing on the Internet, nor protect them against the harmful impact of this exposure on their drinking. Findings presented in this study suggest a need for governments to address this issue by limiting the volume of alcohol marketing in digital media through legislation.

A shortcoming of this study is the measurement of online alcohol marketing exposure (and ownership of alcohol-branded items) by self-reported exposure. These memory-based measures can be strongly affected by the interpretation of the respondent and can be highly correlated to potential confounders, such as past drinking experience (Stacy *et al.*, 2004) in the sense that drinkers may be more familiar with the product and may memorize alcohol advertisements better. Digital tools that automatically identify and register online alcohol advertisements when being encountered avoid the necessity of explicit reporting by the respondent. Unobtrusive measurements such as digital tools, despite their own shortcomings, deserve consideration in future research.

Due to the cross-sectional design, we cannot rule out the possibility that higher (report of) online alcohol marketing exposure is a result of past drinking experience. However, similar effects were found when studying more indirect measures of alcohol marketing exposure. Exposure to televised alcohol advertising was found to be a significant predictor of drinking behavior as well. This alcohol marketing measure was not measured by recall of alcohol advertising, but by recall of television programs (which the authors connected to the number of alcohol commercials broadcasted in these programs). We found a moderate association between exposure to television programs that contained alcohol commercials in the commercial breaks and alcohol use. A similar association was not found for exposure to television programs without alcohol commercials.

## CONCLUSIONS

The cross-sectional data analysis presented suggests that higher exposure to (online) alcohol marketing was found to be associated with drinking behavior (onset of drinking, past 30 days binge drinking and the volume of consumption among already drinkers). This finding was consistent after controlling for confounders such as hours spent using the Internet, demographic and social factors. Consequently, alcohol marketing on the Internet (similar to alcohol marketing practices in other channels) can be seen as a serious but avoidable danger to adolescents' health. The consistency of this effect and the size of the effect among the four European countries seriously raise the demand for legal restrictions of the volume of alcohol marketing in online media in European countries.

## SUPPLEMENTARY MATERIAL

Supplementary material is available at *Alcohol and Alcoholism* online.

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## CONFLICT OF INTEREST STATEMENT

None declared.

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